

JEE MAIN REVISION MASTER: MAGNETISM AND MATTER

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Magnetic Materials (Dia, Para, Ferro) & Hysteresis	11 PYQs	Materials
Moderate	Magnetic Dipole Moment & Earth's Magnetism	7 PYQs	Dipoles & Terrestrial

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Materials (Dia, Para, Ferro) & Hysteresis	$\vec{B} = \mu_0(\vec{H} + \vec{M})$, $\mu_r = 1 + \chi$, $\chi_{\text{para}} = \frac{C}{T}$, $\chi_{\text{ferro}} = \frac{C}{T - T_c}$
Magnetic Dipole Moment & Earth's Magnetism	$\vec{M} = I\vec{A} = NIA$, $\mu_B = \frac{e\hbar}{2m} \approx 9.27 \times 10^{-24} \text{ A m}^2$, $B_H = B\cos\delta$, $\tan\delta = \frac{B_V}{B_H}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Materials (Dia, Para, Ferro) & Hysteresis (High Yield — 11 PYQs)

Core Principle: Classification of magnetic materials, Curie-Weiss law, and hysteresis loop area (energy dissipation).

■ **Caution / JEE Pitfall:** Diamagnetic materials have small negative susceptibility $\chi < 0$ that is strictly independent of temperature.

■ **Speed Trick:** Permanent magnets require high retentivity and high coercivity (e.g. Alnico). Transformer cores require high permeability and low hysteresis loss (soft iron).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substa..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If a paramagnetic bar is brought near a bar magnet, then it is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following statements is true in respect of diamagnetic substances?..." [Ans: N/A]

3.2 Magnetic Dipole Moment & Earth's Magnetism (Moderate Yield — 7 PYQs)

Core Principle: Current loops as magnetic dipoles, orbital electron magnetic moment, and elements of Earth's magnetic field.

■■ **Caution / JEE Pitfall:** At the magnetic equator, dip angle $\delta = 0^\circ$ ($B_V = 0$). At magnetic poles, dip angle $\delta = 90^\circ$ ($B_H = 0$).

■ **Speed Trick:** Apparent dip δ_1, δ_2 in two mutually perpendicular vertical planes: $\cot^2 \delta = \cot^2 \delta_1 + \cot^2 \delta_2$.

Representative Source PYQs:

- **[2023 Shift] Q#3 (MCQ):** "The horizontal component of Earth's magnetic field at a place is $3 \times 10^{-5} \text{ T}$. If the dip at that place is 45° , the result..." [Ans: N/A]
- **[2021 Shift] Q#4 (MCQ):** "Earth's magnetic field always has a horizontal component except at..." [Ans: N/A]
- **[2019 Shift] Q#3 (MCQ):** "A magnetic needle has a magnetic moment of $5 \times 10^{-2} \text{ Am}^2$ and moment of inertia $8 \times 10^{-6} \text{ kgm}^2$. It has a period of osc..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substances with $-1 \leq \chi < 0$ are diamagnetic. II. Substances with $\chi \ll 1$ are paramagnetic. III. Substances with $\chi \gg 1$ are ferromagnetic. IV. Substances with $\mu_r \gg 1$ are ferromagnetic.

- (A) I and II
- (B) III and IV
- (C) II and III
- (D) II and IV

Correct Answer: None

Solution Analysis: The wrong statements are II and III. For magnetic substances, remember these NCERT-based facts: Diamagnetic substances have small negative susceptibility: $-1 \leq \chi < 0$. So statement I is correct. Paramagnetic substances have small positive susceptibility: $0 < \chi \ll 1$. They do not have $\chi \gg 1$. So statement II is wrong. Ferromagnetic substances have very large positive susceptibility: $\chi \gg 1$. They do not have $\chi \ll 1$. So statement III is wrong. For ferromagnetic substances, relative permeability is also very large: $\mu_r \gg 1$. So statement IV is correct. Hence, the wrong statements are: \$

Benchmark Problem 2: KCET 2023 (Q#3)

The horizontal component of Earth's magnetic field at a place is $3 \times 10^{-5} \text{ T}$. If the dip at that place is 45° , the resultant magnetic field at that place is

- (A) $3 \times 10^{-5} \text{ T}$
- (B) $\frac{3}{\sqrt{2}} \times 10^{-5} \text{ T}$
- (C) $\frac{3}{2\sqrt{3}} \times 10^{-5} \text{ T}$
- (D) $3\sqrt{2} \times 10^{-5} \text{ T}$

Correct Answer: None

Solution Analysis: Given, $B_H = 3 \times 10^{-5} \text{ T}$, $\delta = 45^\circ$. From the diagram $\cos \delta = \frac{B_H}{B} \Rightarrow B = \frac{B_H}{\cos \delta} = \frac{3}{\cos 45^\circ} \times 10^{-5} = 3\sqrt{2} \times 10^{-5} \text{ T}$